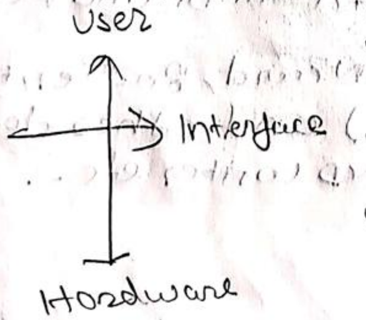
## 1.1 Operating System - Concept, Components of Operating System, operations of OS: Program management, Resource management, Security and Protection.

**Operating System:** (Aatma) (heart-CPU) Concept

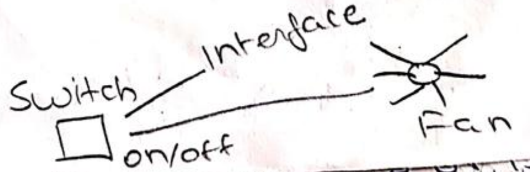
It is a set of programs which controls all the computer resources and provide environment in which user can develop application programs.

An Operating System is a software act as interface between end-users and hardware.

It creates user-friendly environment.




If there is no operating system user can't interact with the hardware. If he wanted to do without O.S. then he have to write program for each and every thing.




~~Operations of OS~~

## Evolution/generation of Operating System.

① First generation (1940-1956) - Vaccum tubes, Plug boards.

* These machines had Vaccum Tubes as a central component.
* machines of the time were so primitive that programs were often entered one bit at a time on rows of mechnical switches (plug boards).
* But the vaccum tubes were having the problem of heating and they required the large space. mainframe
* There was no operating System in this generation.
* Programming languages were unknown (not even Assembly language)
* So all the programming was done in machine language. (Binary) 101010
* All the problems were simple numerical calculations.

② Second Generation (1956-1963) - Transistors, Batch Systems :- System

* In this generation, the vacuum tubes were replaced by transistor.
* So, the problem of size and space is solved.
* The cost was changed dramatically (low)
* Second Generation computers followed Batch System i.e one by one job [program+data=job] were executed in the system.

Batch: collection of job.

* These were called single-stream batch processing systems because programs and data were submitted in groups or batches.



* The Assembly language ... generation.
* The draw back of transistors was leakage current.

P1 P2 → P1 → P1 → I/O

③ Third Generation (1964-1971) - ICs and multiprogramming

* The transistors were replaced by Silicon Chip.
* In this generation, operating system designers the concept of multiprogramming in which several ~~job s are~~ jobs are in main memory at a time; a processor is switched from job to job as needed.
* This meant that processor was not idle while a job was completing its I/O operation. Another job was scheduled on the processor so that its time would not be wasted.
* Users interacted with the third generation computers using keyboard and monitors.

**Q] List any four functions of operating system.**

***(Any four functions - 1 mark each)***

**Ans: The major functions of an operating system are:**

**1. Process Management** – A program in execution is called as Process. OS is responsible for creating, deleting, suspending and resuming the processes.

**2. Memory Management** – OS is responsible for allocation and deallocation of memory to the system programs as well as other programs and data.

**3. File Management –** An OS deals with the **storage**, **creation** and **deletion** of file on various storage devices. It also allows all files to be easily **changed and modified** through the use of text editors.

**4. Device / Resource Management –** Managing devices and resources and allowing the users to **share the resources** such as **CPU time, main memory, secondary storage and input output devices.**

**5. Security and Protection – Securing** the **system** against possible **unauthorized access** to data or any other entity by means of passwords and similar other techniques.

**6. Job Accounting-** Keeping track of **time and resources** used by **various jobs and users.**

**7. Communications –** Providing **interface for the user to communicate** with each other in the same or other system by sharing or transferring data.

1.2 Different types of Operating System: Batch Operating System, Multi-programmed, Time Shared Operating System, Multiprocessor System, Distributed System, Real Time System, Mobile OS (Android OS

## Batch Operating System

The users of Batch operating system do not interact with computer directly.

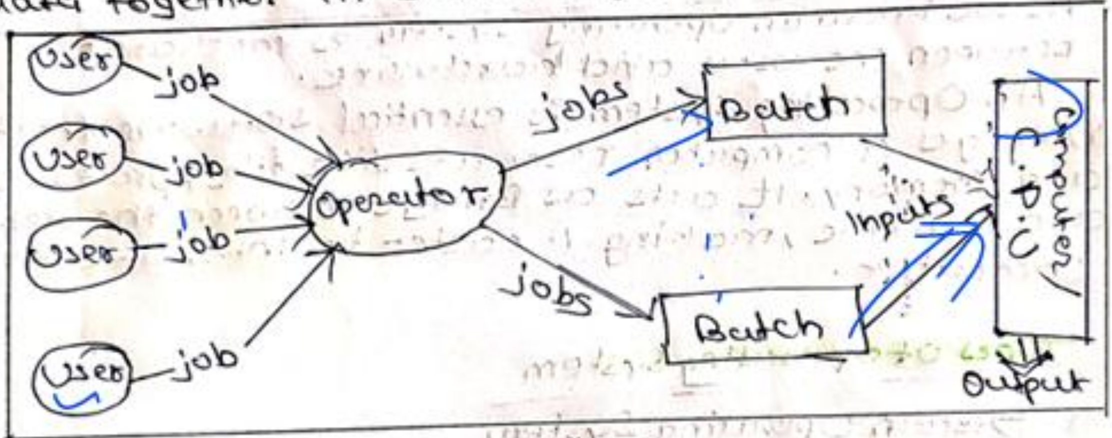
Each user prepares his job on offline device like punch cards and submit it to the computer operator (person).

The Batch operating System will work to ~~Sumit~~ Submit similar kinds of jobs together.

Job : Job is collection of program, I/P data and Control Instruction

In Batch operating System, a batch is a sequence of user jobs.

Here operating system collects program and data together in batch before processing starts.




It follow following activities

1. Operating System defines a job which has predefined sequence of command, programs and data in single unit.
2. operating system keeps a no. of jobs in memory and executes them without any manual information
3. Jobs are processed in order of submission i.e First come first served.
4. ~~After~~ when jobs are ~~com~~ executed the operator seperates the similar kind of jobs in batch.
5. As an input Batchs are given to Computer CPU it executes it in FIFO and process it one batch at a time & provide desired output

Applications: Payroll systems, Bank statements etc.

## Multiprogrammed Operating System

Multiprogramming operating system can be simply illustrated as more than one program present in main memory and one of them kept for in execution.

This is Basically used for better execution of resources.

The operating system keeps several jobs in memory at a time.

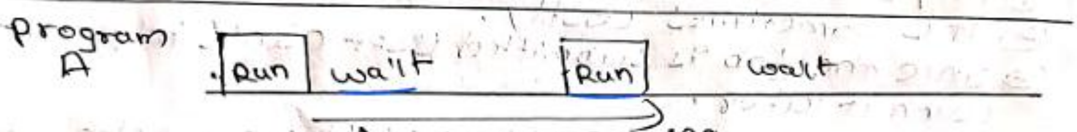
~~The operating systems keeps several job in memory at a time.~~

O.s picks and begin to execute one of job in memory. Eventually the job may have to wait for I/O operation to complete.

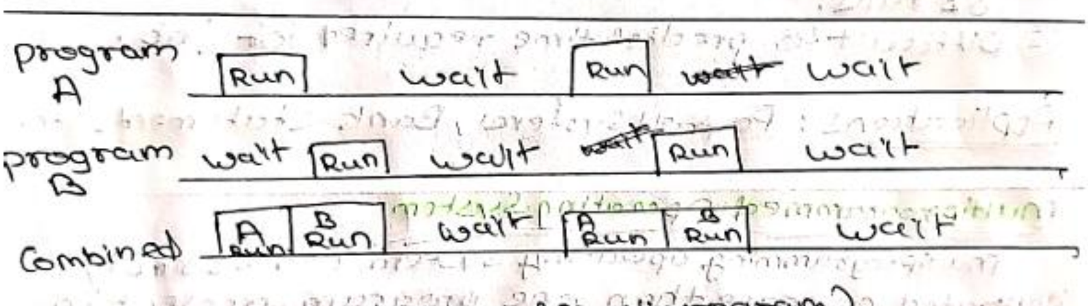
In Non-Programming mode System; CPU will Sit idle.

As long as there is always some job to execute, the CPU will never be idle.

Throughput of system will also increase.

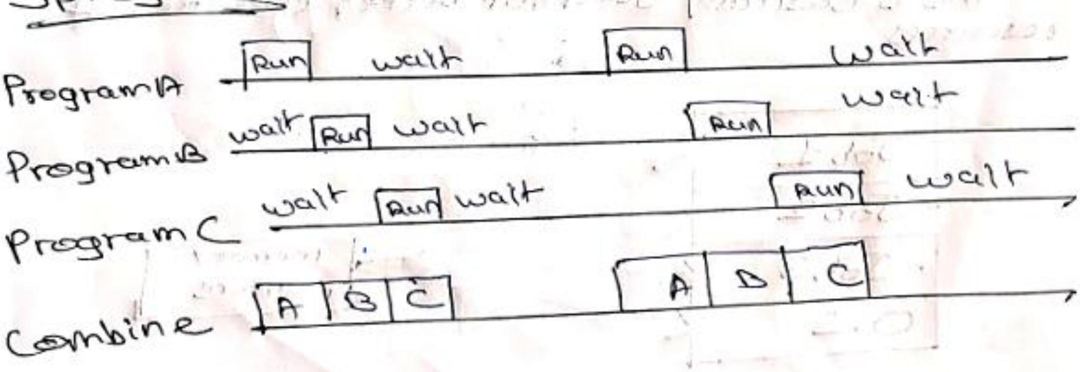



a) Uni programming



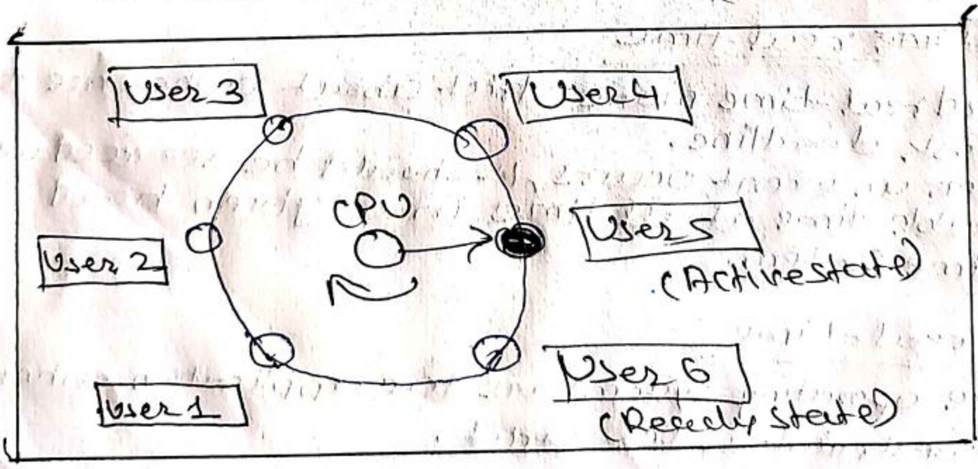

b) two (multiprogram)

3programs



## Time sharing operating system

- Each task is given some time to execute so that all the tasks work smoothly.
- Each user gets the time of CPU as they use a single system.
- These system are also known as multitasking systems.
- The task can be from a single user or different users also.
- The time that each tasks gets to execute is called Quantum.
- After this time interval OS switches over to next task.





### Advantages

- Each task get equal opportunity.
- ~~Fewer chances of duplication of software~~
- CPU ideal time can be reduced.
- Avoids duplication of software.

### Disadvantages

- Reliability problem.
- Data communication problem.
- One must take care of the security and integrity of user programs and data.

Q

a) Multiprocessor System

1) When more than one CPU are user in execution of the processes is known as multiprocessor system.
2) Many CPU are assigned in one cabinet to worked in one cabinet for closed communication.
3) Multiprocessor system are also known as Parallel Systems or tighed coupled systems.
4) It is illustrated as multi-processor which leads to better throughput rate.
5) Here CPU shares Registers and cached memory and connected main memory.

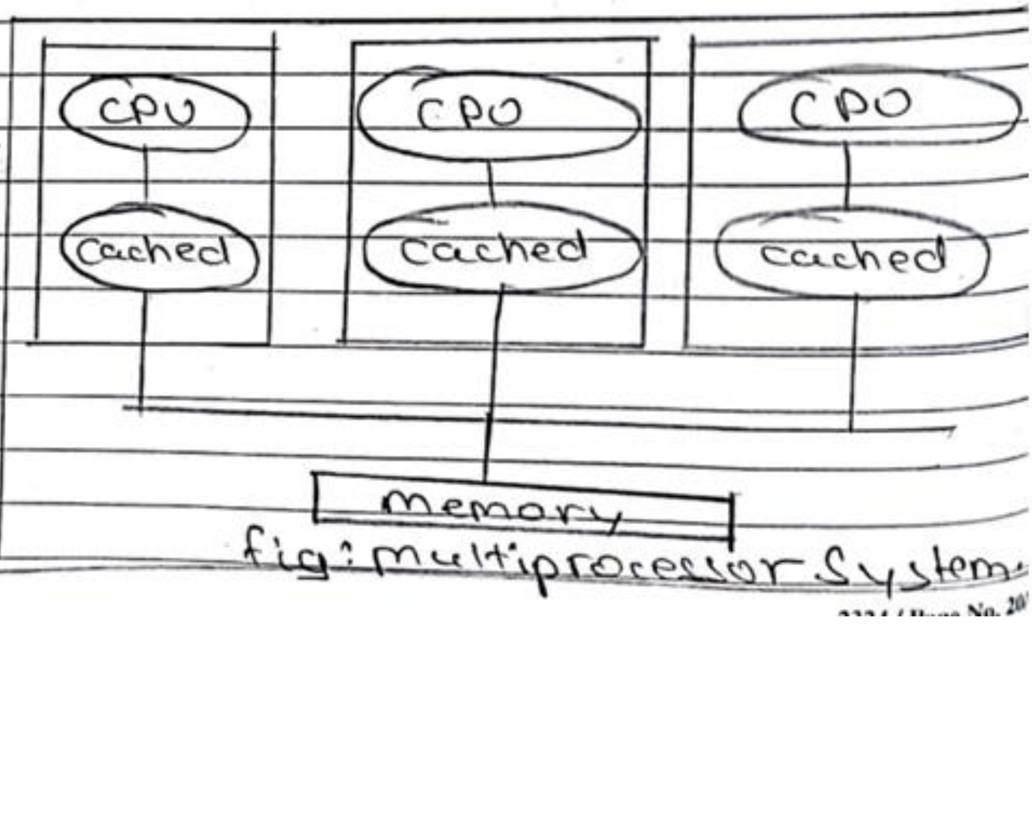



fig: multiprocessor system.

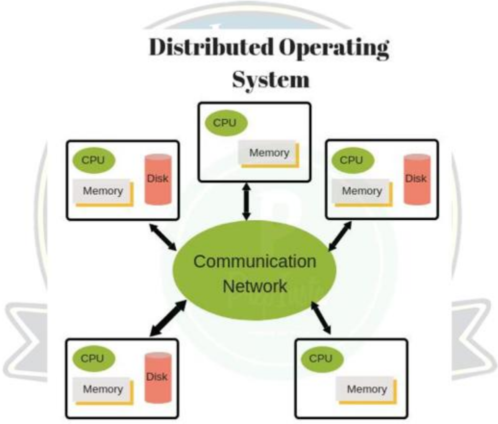
There are two types of multiprocessor System:
1. Symmetric System
2. ASymmetric System.

1. Symmetric System
There is single multiple CPU which are treated equally. No Master-slave relationship.

2. ASymmetric System
There is one main CPU and other CPU are followed or instructed by main. Follows Master-slave Relationship.

5. DISTRIBUTED SYSTEM A distributed system is a network that consists of autonomous computers that are connected using a distribution middleware. They help in sharing different resources and capabilities to provide users with a single coherent network. In this system, processor does not share memory or clock instead each processor has its own memory. Processors in a distributed system may vary in size and function. These processors are referred as sites, nodes, computers, and so on. The nodes communicate with each other through high speed buses or telephone lines by message passing. These systems are also referred as Loosely Coupled System.

These systems are also referred as **Loosely Coupled System.**




**Advantages:**

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## Real-Time Operating Systems

These types of OS serve real time systems. The time interval required to process and respond to inputs are very small.

This time interval is called response time.

Real time systems are used when there are time requirements that are very strict like missile systems, airtraffic control, robots etc....

Types of Real-time OS

Hard Real-time ← OS → Soft Real-time

### Hard ~~time~~ real-time

Hard real-time means strict about adherence to each task deadline.

when an event occurs, it should be serviced within predictable time at all times in a given hard real time system.

### Soft real-time

These operating systems are for application when time constraint is less strict.

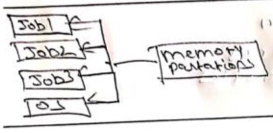
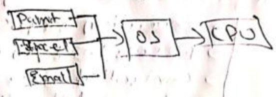
### Applications

↳ Flight Control Systems

↳ Simulations.

| iOS | Android |
|---|---|
| Apple is Developer | ~~Microsoft~~ Google is Developer |
| OS Family - Unix, OS X | OS Family - Linux |
| No, ~~except of~~ widgets except Notification center | Yes, there are widgets |
| Programmed in C, C++, C# | Programmed in C, C++, java. |
| Closed, with open source components | Open source |
| Messaging - iMessage | Messaging - Google Hangouts |
| Launched in 2007 | Launched in 2008 |
| Costly | Cheaper |

| Features | Time sharing | Real time O.S |
|---|---|---|
| Definition | A time sharing system allows several users to use a computer system from various location simultaneously. | A Real time OS completes a task within a specific time. |
| Computer Resources | User may share the resources between users. | Resources are alloted to a process for a fixed amount of time before reallocating to another process. |
| Switching | A switching method/function is available | The switching method/fun^n is not available. |
| modification of program | Any modification may be possible in program | Any modification may not be possible in the program |
| Response | The response is generated within seconds. | The user must get the response within specified time constraint. |
| Process details | It runs the multiple applications simultaneously | It runs only single application simultaneously. |
| Basic | It focus is on responding quickly to a request | It focuses on completing computer task before the deadline. |

| | |
|---|---|
| multiprogramming, multiple processes run concurrently at the same time on a single processor. | multi tasking is when more than one task is executed at single time utilizing multiple CPUs. |
| It is based on the concept of context switching. | It is based on concept of time sharing. |
| It takes more time to execute the processes. | It takes less time to execute the tasks. |
| It utilizes the single CPU for execution of processes. | It utilizes multiple CPU for ~~execu~~ allocation of task. |
| ~~It takes ...~~ more It speed is slow Easy to use | It Speed is fast, Difficult to use |
|  |  |

Lecture 2

1.3 Command line based Operating System: DOS, UNIX GUI based Operating System: WINDOWS, LINUX, MaC OS

| Command Line Based OS | Graphical User Interface OS |
| --- | --- |
| Interaction is by typing commands | Interaction with devices is by graphics and visual components and icons. |
| Commands needs to be memorized | Visual indicators and icons are easy to understand |
| Less memory is required for storage. | more memory is required as visual Components are involved |
| Accuracy is High | Accuracy is Low |
| CLI is difficult to use | GUI is easy to use |
| CLI is faster than GUI | GUI is slower than CLI |
| CLI operating system only needs keyboard. | while GUI needs both mouse & keyboard. |
| CLI apperance can't be changed or modified. | GUI apperance can be changed or modified |
| There are no graphics | graphics are used. |
| Does not avoid spealling mistakes and typing errors | whereas GUI avoid type errors and Spealling mistake. |
| CLI do not use any pointing devices | while GUI uses pointing devices, for selecting and choosing item. |
| In CLI we can obtain high precision | In GUI we can obtain low precision |
| Input is entered on cmd command prompt only. | while GUI, the input can be entered anywher on the screen. |

## Services

- User Interface
- Program Execution
- File system manipulation
- Input/output Operation
- communication
- Resource Allocation
- Error Detection
- Accounting
- security and protection

### 1) User Interface

- User can interact with the system using user-interface service of operating System.
- User interact with Operating System in three ways or interf
  1) Command-line Interface
  2) Graphical user Interface
  3) Batch Based Interface.

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## 2) Program Execution

- Computer or operating system have smaller main-memory and larger secondary-memory
- O.S can't access secondary memory programs they needs to bring in main-memory for execution.
- we not need to see allocation and deallocation O.S takes care of it.

## 3) File System manipulation.

- File System contains various directories & sub-directories which has file in it.
- We User manipulate the ~~thing~~ thing in File System or modification, updation, deletion etc..... comes under this services. Such as creating a file, deleting a file, Renaming a file etc......

## 4) Resource managements.

4) Resource management:

- When process is in "running state" it requires several resources
- Resources such as main memory, file, drivers etc...

P.T.O

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- Operating system does the job of allocating and deallocating resources.





5) Error Detection

- Error is an abnormal condition occurs and system stucks
- It's duty or job of operating system if any error is discovered then inform user via interface.

eg. mouse not working, syntax error etc...

syntax error etc...

## ⑥ Security & Protection.

– User can provide access to limited people in system is Security

04

– Protection is the files should not get corrupted etc...

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... write file, append files, deleting the files, Renaming the file etc.....

It maintains details of files or directories with ~~respect~~ their respective details.

## Communication

The Operating system manages the communication between processes.

One process can communicate with other process with the help of operating System.

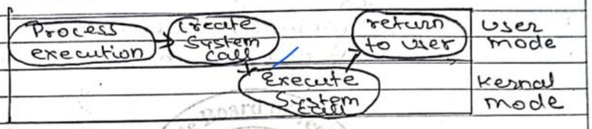
May both process resides in same computer or may resides in other computer

So, one process communicate with another ...

b) System call

System call is call that provide OS services to user via API (Application Programming Interface).

It is the interface between user and kernal.




Following are types of system call :-

1) Process control
2) File management
3) Device management
4) Communication
5) Information maintenance

Types of System calls.

① Process & Job control

A running program needs to be able to halts its execution either normally (end) or abnormally (abort),

If program discovers an error in input & wants to terminate abnormally, it may also want to define error level.

When a process or job is running one program, it might need to start and run another program. This is useful because it allows a system to follow instructions from a user or script to run specific tasks.

(If we start a process or job we should able to control its execution.)

We may also want to terminate a job or process that we created (terminated process). If we find that it is incorrect or no ~~log~~ longer needed we need waiting time to finish execution.

Another set of system calls are helpful in debugging a program.





Examples of Process Control System call

- end, abort
- load, execute
- create process, terminate process
- wait for time, wait event, signal event
- allocate and free memory
- get process attributes, set process attributes.

② File management

System calls which deal with operations related to files fall under this type.

System allows us to create and delete files. For creating and deleting operating system call requires the name of file and other attribute of file.

File attributes such as file type, file size, protection codes etc-----

System access these attributes for performing operation on file and directories.

Once file is ~~op~~ created, we can open it and use it. System allows performing reading, writing or respositioning operation on file.

Examples → create file, delete file
→ open, close

operation on file and directories.

Once file is ~~op~~ created, we can open it and use it. System allows performing reading, writing or repositioning operation on file.

Examples → create file, delete file
→ open, close
→ read, write, reposition
→ get file attributes, set file attributes.

③ Device management

When process in "running" state, it require several resources to execute. These resources include main memory, disk drivers, files and so on.

So system calls are used for asking permission from kernal. If resource is available, it is assigned to process.

Once the resource is allocated to process, process can read, write and reposition device.

→ Request device, ~~relea~~ release device
→ read, write, reposition
→ get device attributes, set device attributes

Scanned with CamScanner

④ Information maintenance

We need to keep information up to date so the system calls helps us to do that.

Transferring information between the user program and operating system requires system call.

System information includes displaying current date and time, the number of current users, the version number of operating system, the amount of free memory or disk space and so on.

Operating system keeps information about all its processes that can be accessed with system call such as get process attributes and set process attributes.

→ get time or date, set time or date
→ get system data, set system data
→ get process, file or device attributes
→ set process, file, or device attributes.

⑤ Communication

Process in system, communicate with each other.

Communication with each other is done by using two models: message passing & shared memory.

For transferring messages, sender process connects itself to receiving process by specifying receiving process name or identity.

Once the communication is over system close connection between communicating processes.

They need to communicate with each other for many reasons like if they need certain resources which is held by another process.

→ create, delete communication connection
→ send, receive messages.
→ transfer status information
→ attach or detach remote devices.

LECTURES 3

1.5 Operating System Components: Process Management, Main Memory Management, File Management, IO Management, Secondary Storage Management

... virtual devices

## Components of Operating System

- Process management
- File management
- I/O System management
- Main memory management
- Secondary storage management

**Process management**

A program is set of instructions.
When CPU is allocated to program it can start its execution.

A program in execution is process. The process which runs simultaneously are managed by process management.

Example, word processing system run by user on PC is a process.

A process needs various system resources CPU, time, memory, File and I/O devices to complete job execution.

These resources can be given to the process when it is created or allocated to it while its running.

The process management activity involves :-

~~Actt~~ Activities performed by process management.

① Provide a mechanism for Deadlock.
② Provide a mechanism for process syncronisation.
③ Provide a mechanism for process monopolisation.
④ Provide a mechanism for process scheduling.
⑤ creating, deleting, termination process among system.
⑥ Divides processes into ~~preemptive~~ higher and lower priorites by premptive scheduling.
⑦ Defines various process states.

Main-memory management

Main memory is an important resource of computer system that needs to be managed by operating system. To execute the program, user needs to keep the program in main memory.

Main memory is volatile.

Every process needs main memory since a process code, stack, heap and data (variables) must all reside in memory.

Main memory is repository of quickly accesible data shared by CPU & I/O devices.

The central processor reads instructions from main memory during the instruction fetch cycle and both reads and write data from main memory during data fetch cycle.

The main memory is one of large storage device that CPU is able to address and access directly.

main memory activities

↳ Allocation of memory to processes.
↳ keep track of memory usuage by the process.
↳ Allocating and deallocating memory space as needed.
↳ Free the memory from processes after completion of execution.

File management

A file is a collected of related information defined by its creator.
Computer can store files on disk (secondary storage) which provide long term storage.
Some examples of storage media are magnetic tape, magnetic tape, magnetic disk and optical disk.
Each of these media has its own properties like speed, capacity, and data transfer rate and access methods. A file system normally organized into the



directories ease their use.
These directories may contain file and other directories.

Activites performed by File management:

① Free space management by techniques Bitmap & Linked List.
② Creating File
③ Deleting a file
④ Write on a file
⑤ Read a file
⑥ Rename a file
⑦ Truncating the file
⑧ Repositining a file
⑨ Accessing a file by - Random



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sequential, swapping method.
⑩ Allocating files by - Linked list allocation, contigoues allocation, index allocation.

- The **backup of files on stable storage media**.

## 4. I/O System Management

**Input / Output device management provides an environment for the better interaction between system and the I/O devices such as printers, scanners, tape drives etc.**

**To interact with I/O devices in an effective manner, the operating system uses some special programs known as device driver.**

A device driver is a specific type of computer software that is developed to allow interaction with hardware devices.

**Activities:**

- Providing interfaces to other system components.
- Managing devices
- Transferring data
- Detecting I/O completion

## 5. Secondary-Storage Management

Secondary storage management

The computer system provides secondary storage to back up main memory.

Secondary storage is required because main memory is too small to accomodate all data and programs, and the data that it holds is lost when power is lost.

Most of programs including compilers, Assemblers, word processors, editors etc.. are stored on disk until loaded into memory.

Secondary storage consist of tapes, drives, disks and other media.

...ible for following activities.

Assemblers, word processors ... loaded ...
Stored on disk until loaded.

Secondary storage consist of tapes, disks and other media.

O.S Responsible for following activities.

- Free space management
- Storage Allocation
- Disk scheduling.

## Use of Operating System tools: